Copernicus Global Land Cover and Tropical Forest Mapping and Monitoring service (LCFM)

PRODUCT USER MANUAL
PAN-TROPICAL TREE COVER DENSITY MAP, 10 M (TCD-10)
VERSION 1

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European Commission
Joint Research Centre
via Enrico Fermi 2749
21027 Ispra (VA)
Italy
https://ec.europa.eu/jrc/en

Document history

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V1_I1.2	25.06.2025	Initial published version
V1_I1.3	19.11.2025	Updated version

Applicable documents

ID	Applicable document
AD-1	Pan-Tropical Tree Cover Density Map 10 m – Algorithm Theoretical Basis Document (ATBD)
AD-2	Pan-Tropical Tree Cover Density Map 10 m – Product Validation Report (PVR)
AD-3	Global Land Cover Map 10 m – Product User Manual (PUM)
AD-4	Global Land Cover Map 10 m – Algorithm Theoretical Basis Document (ATBD)

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1 Executive summary

Monitoring global land cover and tropical forests is critical for addressing global challenges such as deforestation, biodiversity loss, and climate change. These challenges have driven the demand for high-quality, timely, and freely accessible land cover data to inform policies and effectively monitor progress towards the Sustainable Development Goal targets.

In response to this need, the Copernicus Global Land Cover and Tropical Forest Mapping and Monitoring Service (LCFM) has been launched as a new initiative under the Copernicus Land Monitoring Service (CLMS). LCFM focuses on high-resolution mapping and monitoring of global land cover and tropical forests. Led by a consortium headed by VITO, and in collaboration with the European Commission's Joint Research Centre, the service builds upon previous projects like the 100 m Copernicus Global Land Cover layers (Buchhorn et al., 2020), the European Space Agency (ESA) WorldCover (Zanaga et al., 2022), and the REDDCopernicus project (Achard et al., 2020).

The LCFM service not only extends the existing time series (2015–2019) of the 100 m Copernicus Global Land Cover layers, but also enhances this service by delivering a high-resolution (10 m), dynamic global land cover as well as pan-tropical tropical forest monitoring products. Specifically, LCFM provides frequent, sub-annual land surface categories and features, which are consolidated into global annual land cover maps and tropical forest monitoring products at 10 m resolution. This unprecedented level of detail supports better decision-making and enables more effective monitoring of environmental changes.

Importantly, throughout the service's duration, user requirements are continuously assessed and consolidated. In parallel, new user engagement activities are carried out to improve understanding and gather further input, ensuring that LCFM outputs are truly fit for purpose.

Launched in 2024, the LCFM service released its first public datasets in June 2025: the 10 m Global Land Cover Map (LCM-10) for 2020 and the 10 m Pan-Tropical Tree Cover Density (TCD-10) for 2020. Moving forward, the service will broaden its scope by producing similar products for the reference years 2021 through 2026, ensuring a comprehensive and up-to-date resource for global land cover and tropical forest monitoring.

The Pan-Tropical Tree Cover Density product provides information about what percentage of a pixel is covered by tree canopy for the reference year 2020.

This Product User Manual (PUM) is the primary document to provide important information to users before handling the 10 m tropical Tree Cover Density Map. It provides an overview of the product's characteristics, including the algorithm, technical characteristics, and limitations. A summary of the key information about the 10 m Pan-Tropical Tree Cover Density Map (TCD-10) is presented in Table 1.

Table 1: Summary of key information.

Product name 10 m Pan-Tropical Tree Cover Density (TCD-10)
Input data Tree Cover Density Training data (TCD-TD), Sentinel-2 L2A (as pre-processed LSF-ANNUAL-S2, see [AD-4]);
File naming LCFM_TCD-10_V<VERSION>_<YEAR>_<LATLON_TILE_3>_<LAYER>.tif Example: LCFM_TCD-10_V100_2020_N00E030_MAP.tif
File format and product content Single band Cloud Optimized GeoTIFF (COG) files with standard metadata attributes.
Projection and grid information EPSG:4326 with WGS 84 datum and grid resolution of 0.000083333333333° (0.3 arcsec or 1°/12000)
Spatial information The LCFM TCD-10 product covers the geographic areas depicted in Figure 1 situated within Longitude [180°W – 180°E], Latitude [35°N – 35°S] in 3° x 3° tiles
Temporal information Year of acquisition: 2020
Data policies Free and open access
Data access https://documentation.dataspace.copernicus.eu/Data/ComplementaryData/CLMS.html#global-dynamic-land-cover
Contact Copernicus Land Monitoring Service – Service desk

2 Scope of the document

2.1 Scope and objectives

This Product User Manual (PUM) is the primary document to provide important information to users before handling the 10 m Pan-Tropical Tree Cover Density Map. It provides an overview of the product's characteristics, including the algorithm, technical specifications, known limitations as well as guidance on accessing the products, terms of use, and technical support.

2.2 Content of the document

This document is structured as follows:

- Chapter 3 summarizes the retrieval methodology.
- Chapter 4 describes the technical properties of the product.
- Chapter 5 summarizes the known limitations.
- Chapter 6 provides more details on how to access the products.
- Chapter 7 summarizes the terms of use and product technical support.

3 Algorithm description

3.1 Overview

The production of TCD-10 for the reference year 2020 is built on the methodology and workflow initially developed for the production of pan-European Forest layers (EEA, 2021). This approach has been significantly improved for covering the tropical areas and tested during the LCFM Ramp-up Phase. The overall workflow combines reference samples from 388 geographic strata and image features in form of annual Land Surface Features derived from Sentinel-2 (e.g. percentiles, medians; referred to as LSF-ANNUAL-S2). The reference data for TCD modelling, the TCD training database (TCD-TD), is derived from very high-resolution image data and provides reference TCD values at 10m spatial resolution.

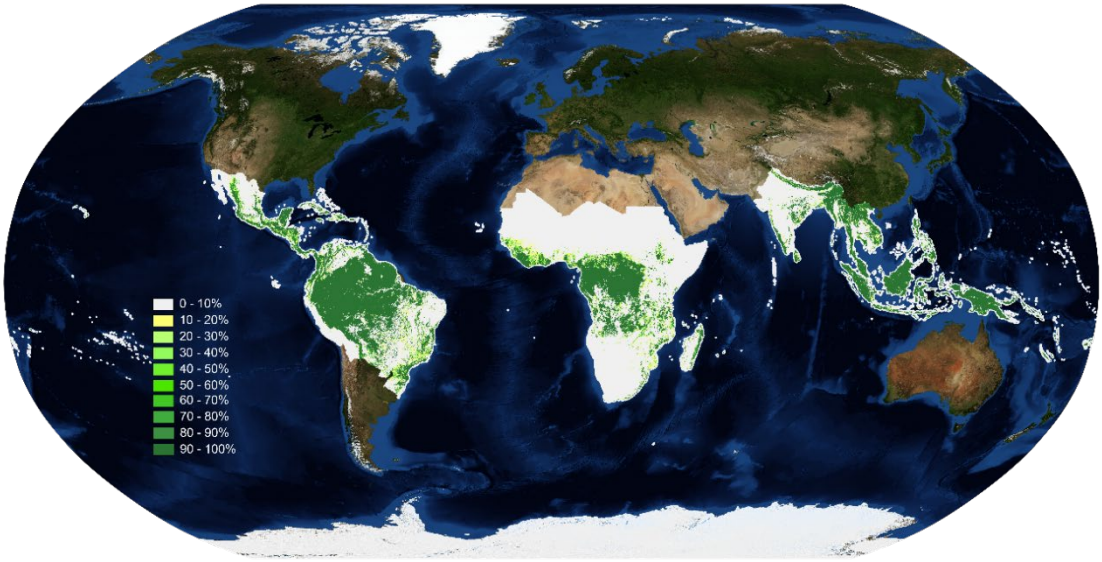


Figure 1: Overview of the Tree Cover Density (TCD-10) product for the pan-tropical area of interest.

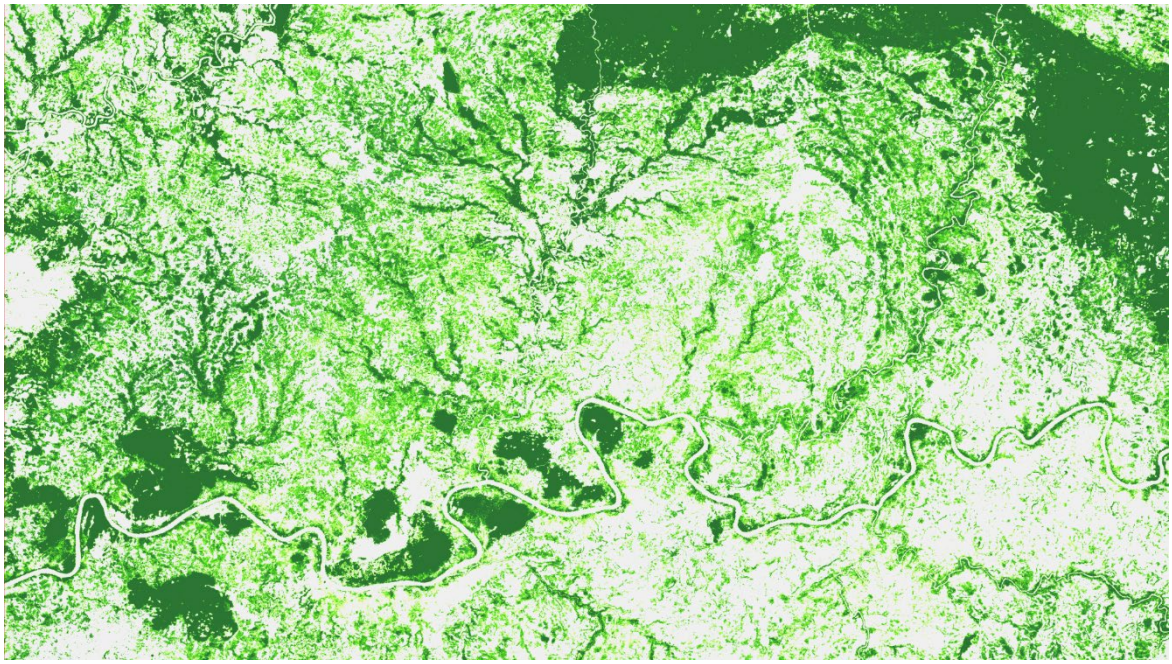


Figure 2: Detailed view of the Tree Cover Density (TCD-10) product in the Banghiang River region, Laos.

The intention of using temporal aggregates (i.e. LSF-ANNUAL-S2) from Sentinel-2 time series for TCD modelling is to represent the variability of spectral signals across all cloud-free observations in a given year. During the ramp-up phase and in close collaboration with the LCFM product Land Cover Map at 10 m spatial resolution (LCM-10), the LSF-ANNUAL-S2 features were defined as percentile aggregates of the Sentinel-2 bands B02, B03, B04, B08, B11, and B12 as well as Normalized Difference Vegetation Index (NDVI) values. This was done to maximize synergies in usefulness of the pre-computed features for both global land cover monitoring and tropical forest monitoring.

Combining the reference observations (TCD-TD) with the LSF-ANNUAL-S2, a machine learning (ML) algorithm was trained to model the relationship between the spectral reflectance signals recorded from Sentinel-2 and Tree Cover Density. This ML model and the entire pan-tropical image database of LSF-Annual imagery were used to derive an area-wide TCD-10 product for the entire tropical area of interest. Internally, the algorithmic approach is designed as a regression algorithm, i.e. a continuous TCD value between 0 % and 100 % is estimated.

However, to reduce the local impact of signal noise from various sources and increase stability, the final product is provided in classes of 10% ranges (see section 4.3.1).

3.2 Input data

The TCD-10 product is generated based on TCD specific training data and aggregates of Copernicus Sentinel-2 L2A data (i.e. LSF-ANNUAL-S2).

3.2.1 Training data

The TCD-10 product was trained using a training dataset of ~17.6 million reference points representative of the spatial distribution and ecosystems across the pan-tropical region. For more technical details on the training data strategy and methodology, please refer to the TCD-10 Algorithm Theoretical Basis Document [AD-1]. The training dataset was derived from over 7,000 Primary Reference Units (PRUs (see Figure 3) each of a 500 m × 500 m area aligned to the 10 m UTM grid of the Sentinel-2 input. Each PRU contains 2,500 10 m pixels, ensuring direct compatibility with the target resolution.

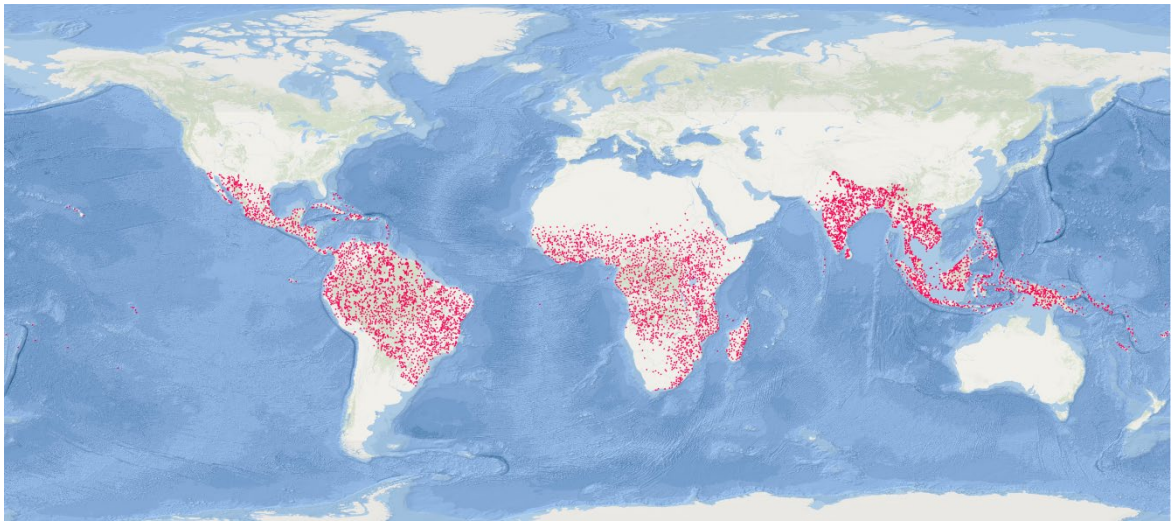


Figure 3: Pan-tropical TCD Training Data PRU sites. Basemap: © Esri World Ocean.

PRUs were distributed across 388 Bio-Geographic Regions (BGRs) generated by segmenting Sentinel-3 NDVI composites from 2020. The eight strata derived in order to obtain a rough understanding of the underlying terrain types per BGR helped to obtain diversity within the sampling strategy.

Very high-resolution reference imagery from 2020 (and 2021, where necessary) was used to derive tree canopy masks. A sampling tool developed for tree cover density training data collection that combines image segmentation and Machine Learning (ML) was used for this purpose (see Figure 4). Using these masks, the percentage of canopy coverage was derived using zonal statistics for each Sentinel-2 pixel of the PRUs.

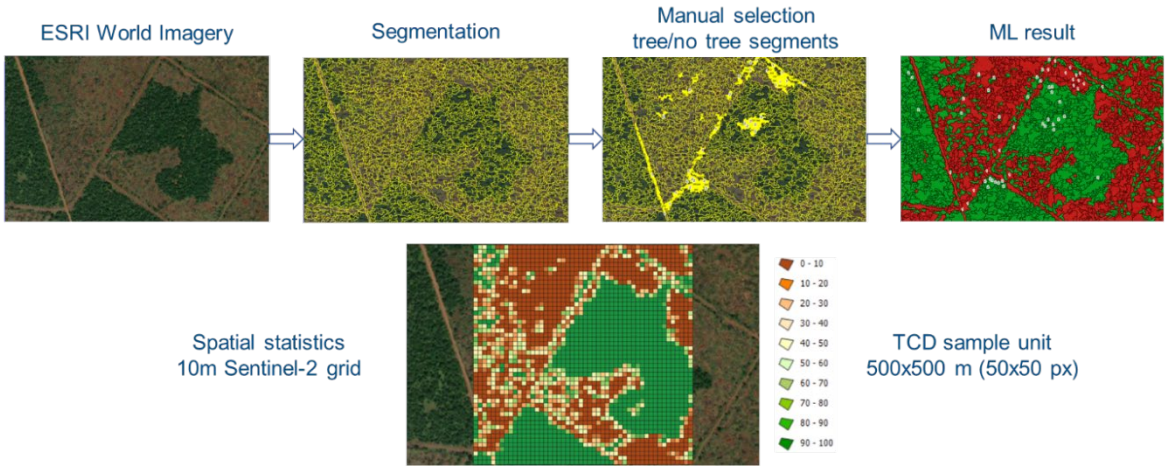


Figure 4: TCD training data collection workflow.

3.2.2 LSF-ANNUAL-S2

The pan-tropical TCD-10 product is based on an annual aggregate product of Sentinel-2 multispectral imagery, i.e. annual Land Surface Features from Sentinel-2 (LSF-ANNUAL-S2) for the year 2020. For more detailed information on the LSF-ANNUAL-S2, please refer to the relevant LCM-10 PUM [AD-3] and ATBD [AD-4].

Sentinel L2A Collection-1 data was downloaded/processed from the Copernicus Data Space Ecosystem (CDSE). Quality checks were performed by the European Space Agency (ESA), resulting in quality flags in the product metadata. Upon ingestion, products containing defective data were discarded. The next two subsections contain the details of the processing steps performed on this dataset.

3.2.2.1 Land Occlusion Score (LOS): an algorithm for Sentinel-2 L2A clouds detection

Copernicus Sentinel-2 L2A products are distributed by ESA with a 20 m scene classification (SCL) layer. This layer includes a classification of cloudy pixels. Cloud detection is sometimes affected by omissions (cloudy observations in reality, but no cloud masking set) which can cause artefacts in the derived annual features. To improve the quality of input features, a custom clouds detection algorithm, called Land Occlusion Score (LOS), was developed using the CloudSEN12+ training dataset (Aybar et al., 2024). The Land Occlusion Score product is produced as a set of four probabilities (surface, clouds, shadows, snow) and is characterized by a continuous distribution of the target variables which are combined to produce an occlusion score at 10 m resolution. The CloudSEN12+ dataset was extended to include additional observations containing snow, one of the target occlusion probabilities. This score is the basis for the following compositing step. For more details on the Land Occlusion Score algorithm, the reader is referred to the LCM-10 ATBD [AD-4].

3.2.2.2 Sentinel-2 L2A features computation

Next, Sentinel-2 L2A features are obtained through the following steps:

- The Sentinel-2 L2A catalog is queried for the period, yielding a set of L2A products.
- Bands B02, B03, B04, B08, B11, B12 of these products are loaded.
- LOS probabilities are loaded for the same L2A products at 60 m and resampled at 10 m.
- Quality weighted compositing: LOS probabilities are combined to derive a compositing weight. The weight is proportional to the estimated quality of the pixel. To account for errors in this estimation and model uncertainties, pixels that are in a 10% quality range of the best selected pixel are merged with a weighted mean, where the weight is proportional to their relative quality.
- Quality weighted reconstruction: After generating the composited time series, residual invalid observations must be removed and missing values filled. Each composite is produced with an additional band, the max validity score for each pixel. This score indicates how "good" the leading pixel was. The timeseries is then re-interpolated to replace lower quality values, reconstructing a cleaned and gap-filled timeseries.
- The obtained timeseries is used to compute the (10th, 25th, 50th, 75th, 90th) percentiles of bands B02, B03, B04, B08, B11, B12 and the NDVI index, computed as the timeseries of (B08 – B04) / (B08 + B04)

For more details on each of these steps, the reader is referred to the LCM-10 ATBD [AD-4].

Finally, 35 Land Surface Features from S2 (LSF-ANNUAL-S2) are obtained from the starting Sentinel-2 L2A input observations and LOS scores, providing the core set of features for the LCM-10 classification.



Figure 5: RGB composite of three Land Surface Features near Virunga National Park (0°N, 29°E) derived from Copernicus Sentinel-2 L2A data showing the 50th percentile of B04, B03 and B02, respectively.

3.3 TCD modelling

In order to train a TCD model, a tabular feature space was generated by spatially overlaying the reference samples from TCD-TD with Sentinel-2-based LSF-Annual image features, linking the

target variable (TCD) with corresponding explanatory image features (see Figure 6). From this tabular data structure, a ML algorithm can be trained to identify the connections between image signal and TCD values. For the creation of the LCFM TCD-10 product, the gradient boosted tree-based ensemble algorithm CatBoost (Prokhorenkova et al., 2018) was employed. This specific algorithm was used for its scalability in large-scale deployments as well as the robust performance in large datasets. The model was found to outperform manually calibrated multivariate regression models which were previously considered as the gold-standard for the TCD estimation.

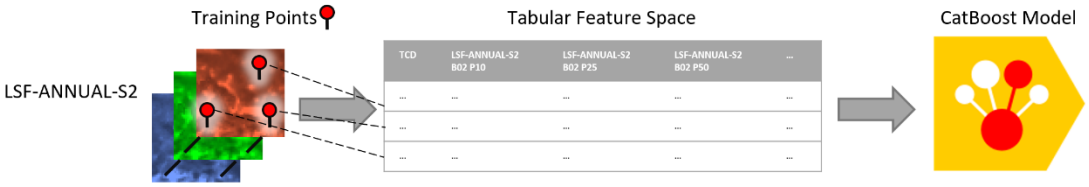


Figure 6: TCD-10 model training workflow.

In practice, the ca. 17.6 million reference sample pixels were employed in a 5-fold cross-validation scheme grouped by location and TCD value range. That is, cross validations folds were created avoiding spatial sample contamination of similar TCD values from the same location. Therefore, groups were build based on the PRU identifier and on two TCD value classes (TCD < 50 %, and TCD >= 50 %). Effectively, whenever a sample from one sample group (e.g. PRU ID 42, TCD < 50 %) was within one fold, no samples from the same group were in any other folds.

The cross validation served two purposes. First, as a boosted tree-based algorithm, CatBoost requires one fold of the data to prevent overfitting of the algorithm. Second, an independent test fold was used to ensure minimal variance from the combination of training and evaluation samples during the training process. A model was trained with a maximum of 1500 iterations. Early stopping after 250 iterations was applied to halt training once performance plateaued, minimizing unnecessary computation and avoiding overfitting.

The trained model was then applied to the set of image tiles spanning the pan-tropical AOI resulting in an initial raw prediction. This raw product (outcome of the model) was further refined using the following post-processing steps:

- **Harmonisation** - Application of a tree probability mask: using the probabilities for the classes “tree cover” and “mangroves” from the product LCFM LCM-10, the TCD-10 product was harmonised with LCM-10 for pixels with low probabilities (for more details, the reader is referred to the TCD-10 ATBD [AD-1]).
- **Mosaicking -** Transformation from UTM projected coordinates to geographic WGS84 decimal degree coordinates: for publication, all products were transformed from UTM coordinates (EPSG: 326xx and 327xx) to the WGS84 coordinate reference system (EPSG: 4326). Therein, adjacent UTM tiles were blended and mosaicked into 3° × 3° tiles. Simultaneously, the product was masked using the pan-tropical AOI as defined by countries borders from the Eurostat GISCO country layer. Along ocean borders, a 30 km buffer was introduced.
- **Binning of TCD values:** Next, the raw continuous TCD values were binned into discrete values of 10 %, 20 %, …, 90 %, 100 % by rounding up to the next value divisible by 10. Thus, the final product has 10 binned classes of TCD cover, with two additional flag values: 254 “Unclassifiable” and 255 “No Data”.

- **Assigning metadata and conversion to Cloud Optimized GeoTiff (COG):** In a last step, the files were assigned with the appropriate metadata and converted to the image file format of Cloud Optimized GeoTiff (COG).

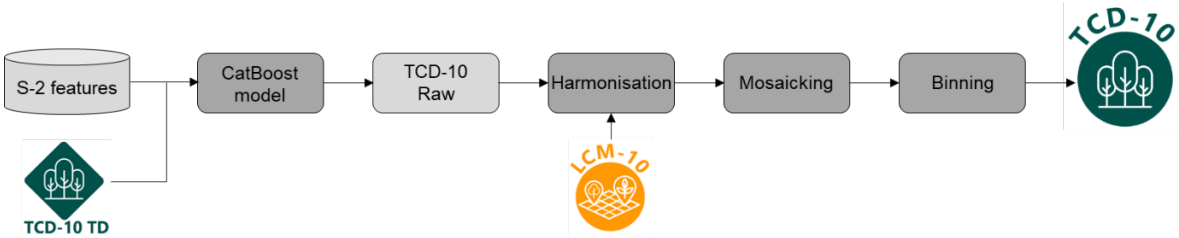


Figure 7: TCD-10 high-level workflow.

3.4 Differences with the previous version

This is the first version of the LCFM TCD-10 product.

4 Product description

The 10 m pan-tropical Tree Cover Map (TCD-10) is distributed in 3° × 3° tiles (Figure 8), 887 tiles in total. Each tile is delivered as a Cloud Optimized GeoTIFF (COG) file in the EPSG:4326, geographic latitude/longitude coordinate reference system (CRS).

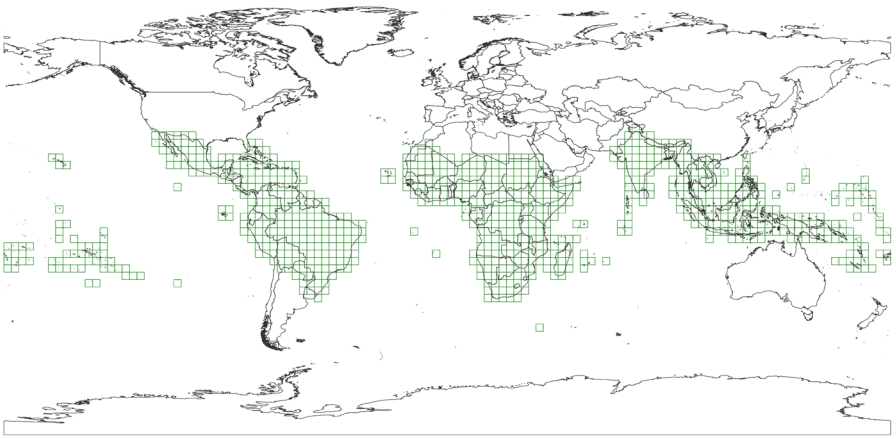


Figure 8: Scheme of the 3° × 3° tiles for the pan-tropical area for TCD-10

4.1 File naming

The LCFM TCD-10 product GeoTIFF files follow this naming standard:

LCFM_TCD-10_V<VERSION>_<YEAR>_<LATLON_TILE_3>_MAP.tif

where:

- <VERSION> shows the processing line version denoted as used to generate the products. The version is denoted as <M><m><r> (e.g. 201), with <M> representing the major version (e.g. V2), <m> the minor version and <r> the production run number (Table 2).
- <YEAR> indicates the product reference year, expressed as YYYY.
- <LATLON_TILE_3> indicates the spatial coverage of the file, composed of the 3-digit longitude and 2-digit latitude of the bottom-left corner of the 3° × 3° tile (e.g. S03E030 covering the area with latitude [-3°, 0°] and longitude [30°, 33°]).

Please note that the file naming might change in a future version, to be consistent with other CLMS products available on CDSE.

Table 2: Explanation in version numbering and recommendations for using efficiently the products

Versions	Differences	User recommendation
Major	Significant change to the algorithm.	Do not mix various major versions in the same applications, unless it is otherwise stated

Minor	Minor changes in the algorithm	Can be mixed in the same applications, but require attention or modest modifications
Run	Fixes to bugs and minor issues, for instance when a reprocessing is necessary due to degraded ancillary data. Later run automatically replaces former.	Consider it as a drop-in replacement

4.2 File format

The 10 m pan-tropical Tree Cover Density Map (TCD-10) is provided as fully compliant single band Cloud Optimized GeoTIFF (COG) files with standard metadata attributes.

4.3 Product content

4.3.1 Product layers

The TCD-10 product consists of a single layer, see Table 3. The details of the layer are described in the sections below.

Table 3: TCD-10 product layer.

Layer name	Content	Number of bands	Data type	No data value	Scale Offset
MAP	Tree Cover Density Map at 10 m resolution	1	UINT8	255	-

MAP

The TCD-10 MAP layer provides the tree canopy density in terms of projective tree cover in percent per pixel for 2020 reference year in a discrete range between 0% to 100%. The continuous TCD has been masked using the tree and mangrove class probabilities from the LCFM LCM-10 product. The product has been binned into 10 bins. Each bin covers a range of ten (10) percentage points of TCD, up to a maximum of 100%. The TCD-10 MAP product is hence a discrete classification map, providing 10 classes (see Table 4).

Table 4: Coding of the Tree Cover Density layer and definition of the bins.

Map code	Tree Cover Percentage	Definition	Colour
10	> = 0% & < =10%	Any geographic area dominated by trees with a cover between 0 % and up to and including 10%.	
20	> 10% & <= 20%	Any geographic area dominated by trees with a cover above 10% and up to and including 20%.	
30	> 20% & <= 30%	Any geographic area dominated by trees with a cover above 20% and up to and including 30%.	
40	> 30% & <= 40%	Any geographic area dominated by trees with a cover above 30% and up to and including 40%.	
50	> 40% & <= 50%	Any geographic area dominated by trees with a cover above 40% and up to and including 50%.	
60	> 50% & <= 60%	Any geographic area dominated by trees with a cover above 50% and up to and including 60%.	
70	> 60% & <= 70%	Any geographic area dominated by trees with a cover above 60% and up to and including 70%.	
80	> 70% & <= 80%	Any geographic area dominated by trees with a cover above 70% and up to and including 80%.	
90	> 80% & <= 90%	Any geographic area dominated by trees with a cover above 80% and up to and including 90%.	
100	> 90% & <= 100%	Any geographic area dominated by trees with a cover above 90% and up to and including 100%.	
254	Unclassifiable	Unclassifiable due to data gaps e.g. persistent cloud cover	
255	No Data	No Data (out of AOI)	transparent

4.3.2 Metadata attributes

The GeoTIFF files provide the metadata attributes as key value pairs:

- on the file-level (Table 5);
- on the band-level, with an example of values given for the discrete classification layer (Table 6).

Table 5: Description of GeoTIFF file attributes.

Attribute	Description	Example(s)
copyright	Copyright notice	LCFM project 2020 / Contains modified Copernicus Sentinel data (2020) processed by LCFM consortium
creation_time	File creation timestamp	2025-05-17T11:17:18Z
description	A description of the contents of the file	Annual Tree Cover Density at 10 m resolution
doi	Digital Object Identifier	https://doi.org/10.2909/59cc02d6-ddfe-4820-83cb-345205eacec5
layer	Layer name	MAP
legend	Legend classes and names	10 : 0% – 10% TCD 20 : 11% – 20% TCD 30 : 21% – 30% TCD 40 : 31% – 40% TCD 50 : 41% – 50% TCD 60 : 51% – 60% TCD 70 : 61% – 70% TCD 80 : 71% – 80% TCD 90 : 81% – 90% TCD 100 : 91% – 100% TCD 254 : Unclassifiable 255: No Data
license	License information	CC-BY 4.0 - https://creativecommons.org/licenses/by/4.0/
producer	Institution that produced the product	GAF AG
product_crs	Coordinate Reference System of the product	EPSG:4326
product_grid	Product grid	3° x 3° tiling grid
mission	Copernicus mission	SENTINEL-2
platform	SENTINEL-2 satellites	SENTINEL-2A, SENTINEL-2B
sensor	Sentinel-2 sensor	MSI
product_tile	Tile from product_grid	N00E015
product_type	Type of product	Categorical
product_name	Name of the product	TCD-10
references	Web reference with more product information	https://land.copernicus.eu
time_end	End date and time of the temporal coverage of the input data	2020-12-31T23:59:59Z
time_start	Start date and time of the temporal coverage of the input data	2020-01-01T00:00:00Z
version	Version of the product	V100
AREA_OR_POINT	Area or point	Area

Table 6: Description of GeoTIFF band attributes.

Attribute	Description	Example for TCD-10 MAP layer
Description	Band name	MAP
NoData Value	Single value, outside of valid range, used to represent missing or undefined data	255
Offset	Offset to apply to the values	NA
Scale	Scale to apply to the values	NA
Colour Table	Colour representation information	RGB with 256 entries

4.4 Product characteristics

4.4.1 Projection and grid information

The LCFM TCD-10 product is delivered in a regular latitude/longitude grid of 3° × 3° tiles (EPSG:4326). The resolution of the grid is 0.3 arcsec (1°/12,000) or approximately 10 m at the equator. The <LATLON_TILE_3> indicated in the file name refers to the spatial coverage of the file, composed of the 3-digit longitude and 2-digit latitude of the bottom-left corner of the 3° × 3° tile.

4.4.2 Spatial information

The LCFM TCD-10 product covers the geographic areas depicted in Figure 1 (between 36°N to 48°S). The products are provided in 3° × 3° tiles with 36,000 x 36,000 pixels each.

4.4.3 Temporal information

The LCFM TCD-10 product represents the Tree Cover Density (TCD) for the reference year 2020 (from 1 January to 31 December).

4.4.4 Thematic information

Tree Cover Density is defined as the vertical projection of tree crowns to a horizontal earth's surface and provides pixel-based tree cover information on the proportional crown coverage per pixel.

Trees are defined as perennial woody plants characterized by a single, self-supporting main stem or trunk that supports branches and leaves in most species. They have a minimum height of at least five meters (≥5 m). This definition encompasses all types of trees, including those in natural or managed forests; lines and shelterbelts along infrastructure features and agricultural fields; scattered trees in agricultural and natural landscapes; tree plantations (such as fruit orchards and palm plantations); and trees in parks, gardens, and around buildings.

Table 7: Thematic elements defining the tree cover for TCD-10.

Elements **Included**	Elements **Excluded**
- Trees in natural forests - Trees in managed forests - Trees outside forests (lines and shelterbelts along infrastructure features and agricultural fields; scattered trees in agricultural & natural landscapes) - Orchards, olive groves, cocoa, oil palm, fruit and other tree plantations - Partially damaged trees caused by wind, fire, insects, and other biotic and abiotic agents	- Shrubland - Open areas within forests - Short woody crops such as tea and coffee - Vineyards - Tall herbaceous vegetation such as bamboo, banana, cacti, sugarcane - Dead trees caused by wind, fire, insects, and other biotic and abiotic agents - Tree nurseries

5 Known limitations

Below we list a series of known limitations of the current TCD-10 MAP layer:

Artefacts / Data gaps

- Data gaps and cloud artefacts are visible in areas where frequent cloud cover impacts the input time features and cloud masking could not sufficiently eliminate the effects.
- S2 tile edges and swath borders can be visible in some areas in the input features and therefore potentially in the TCD-10 MAP layer.

Limitations caused by the applied tree mask

- Mismatch between Tree Mask and TCD-10 MAP layer during product harmonization: masking of areas where TCD-10 actually does detect tree cover, and unmasked areas where TCD-10 does not detect trees lead in some areas to a perceived underestimation of tree cover.

Limitations caused by the TCD model

- Overestimation and underestimation of TCD values can appear in certain areas.

Region	**Description of the issue**
Colombia (North West)	Underestimation of TCD values (cloud artefacts)
Ecuador (West)	Underestimation of TCD values
Madagascar (East)	Overestimation of TCD values
Paraguay (North)	Underestimation of TCD values
Venezuela (West)	Underestimation of TCD values (cloud artefacts)

6 Product access

The LCFM products are integrated into the Copernicus Data Space Ecosystem (CDSE) and are available through S3, OData, and the Copernicus Browser.

6.1 Copernicus browser

The Copernicus browser accessible through https://browser.dataspace.copernicus.eu provides functionality to **access**, **visualize** and **analyse** the TCD-10 products. Furthermore, it also allows users to **compare** the map with other products, such as Sentinel-2 quarterly composites. Finally, users can also use the browser to **download** the TCD-10 products.

General documentation on the browser can be found here:
https://documentation.dataspace.copernicus.eu/Applications/Browser.html

To access TCD-10 specifically, follow this browser link or open the collection from the browser landing page:

- In the 'Default' configuration, under 'Data Collections' on the left panel, select 'Copernicus Land Monitoring Service' from the dropdown menu.
- Select 'CLMS Land Cover and Land Use Mapping'
- Select 'Dynamic Land Cover'
- At the top under 'Date: Single', select 'Show latest date'
- Select 'TCD Pan-tropical 10m Yearly V1'

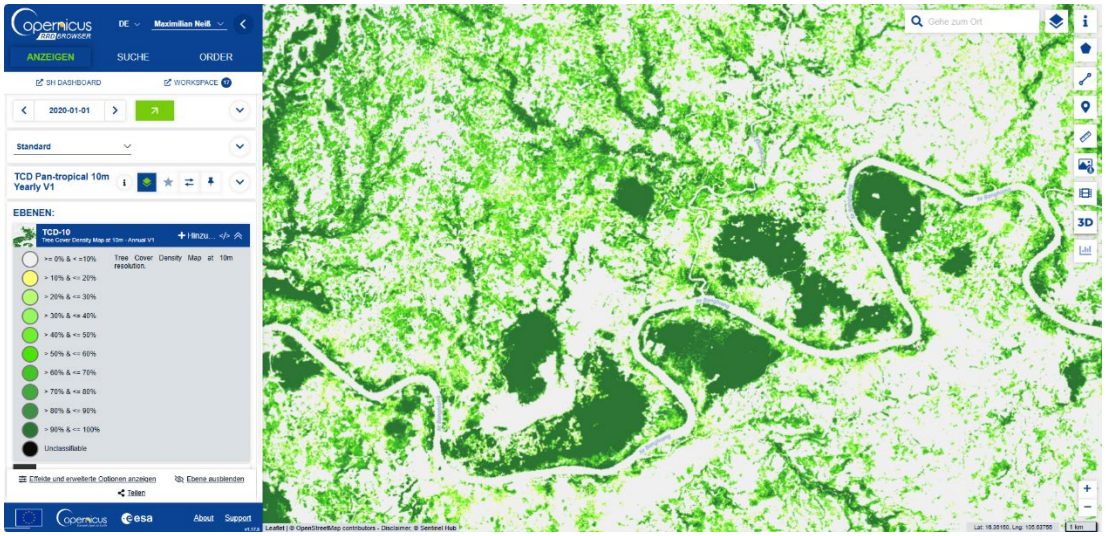


Figure 9: Copernicus browser showing the TCD-10 map.

6.2 Simple Storage Service (S3)

The products are hosted on the EODATA bucket hosted by CloudFerro. To access, please refer to this web page:
https://documentation.dataspace.copernicus.eu/APIs/S3.html

The path to access the products is:

s3://eodata/CLMS/landcover_landuse/dynamic_land_cover/tcd_pantropical_10m_yearly_v1/2020/01/01/

6.3 OData

The Tropical Tree Cover Density (TCD-10)) products are accessible via the OData interface, using the dataset identifier **tcd_pantropical_10m_yearly_v1**.

OData (Open Data Protocol) is a standardized protocol for querying and updating data via RESTful APIs. The Copernicus Data Space Ecosystem supports OData to allow users to filter and access metadata and product resources programmatically.

For an overview and detailed explanation of how to interact with the OData interface, please refer to the official CDSE OData documentation:
https://documentation.dataspace.copernicus.eu/APIs/OData.html

To retrieve all products associated with the tcd_pantropical_10m_yearly_v1 dataset, use the following query:
https://catalogue.dataspace.copernicus.eu/odata/v1/Products?$filter=Attributes/OData.CSC.StringAttribute/any(att:att/Name%20eq%20%27datasetIdentifier%27%20and%20att/OData.CSC.StringAttribute/Value%20eq%20%27tcd_pantropical_10m_yearly_v1%27)

This will return the first 20 products, for the other ones use the link from @odata.nextLink.

To access a specific product by its filename (e.g., a Cloud Optimized GeoTIFF for the 2020 map of tile N51W066), you can use a query like this:
https://catalogue.dataspace.copernicus.eu/odata/v1/Products?$filter=Name%20eq%20%27LCFM_TCD-10_V100_2020_N03E003_cog%27&$expand=Attributes

7 Terms of use and product technical support

7.1 Terms of use

The Terms of Use for the product(s) described in this document acknowledge the following: Free, full and open access to the products and services of the Copernicus Land Monitoring Service is made on the conditions that:

1. When distributing or communicating Copernicus Land Monitoring Service products and services (data, software scripts, web services, user and methodological documentation and similar) to the public, users shall inform the public of the source of these products and services and shall acknowledge that the Copernicus Land Monitoring Service products and services were produced "with funding by the European Union".
2. Where the Copernicus Land Monitoring Service products and services have been adapted or modified by the user, the user shall clearly state this.
3. Users shall make sure not to convey the impression to the public that the user's activities are officially endorsed by the European Union.

The user has all intellectual property rights to the products he/she has created based on the Copernicus Land Monitoring Service products and services. Consult Data policy — Copernicus Land Monitoring Service for further details.

7.2 Citation

When planning a publication (scientific, commercial, etc.), it shall explicitly mention:

> **"This publication has been prepared using European Union's Copernicus Land Monitoring Service information; https://doi.org/10.2909/59cc02d6-ddfe-4820-83cb-345205eacec5."**

When developing a product or service using the products or services of the Copernicus Land Monitoring Service, it shall explicitly mention:

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When redistributing a part of the Copernicus Land Monitoring Service (product, dataset, documentation, picture, web service, etc.), it shall explicitly mention:

> **"European Union's Copernicus Land Monitoring Service information; https://doi.org/10.2909/59cc02d6-ddfe-4820-83cb-345205eacec5. Contains modified Copernicus Sentinel data (2020)."**

Consult Data policy — Copernicus Land Monitoring Service for further details.

7.3 Product technical support

Product technical support is provided by the product custodian through Copernicus Land Monitoring Service – Service desk. Product technical support does not include software specific user support or general GIS or remote sensing support.

More information on the products can be found on the Copernicus Land Monitoring Service website (https://land.copernicus.eu/).

List of abbreviations

Abbreviation	Name
AD	Applicable Document
ATBD	Algorithm Theoretical Basis Document
AWS	Amazon Web Services
BGR	Bio-Geographic Region
CLI	Command Line Interface
CDSE	Copernicus Data Space Ecosystem
CLMS	Copernicus Land Monitoring Service
CRS	Coordinate Reference System
COG	Cloud Optimized GeoTIFF
EO	Earth Observation
ESA	European Space Agency
EU	European Union
GUI	Graphical User Interface
JRC	Joint Research Centre
LCFM	Copernicus Global Land Cover and Tropical Forestry Mapping and Monitoring Service
LCM	Land Cover Map
LOS	Land Occlusion Score
LSF	Land Surface Features
ML	Machine Learning
NDVI	Normalized Difference Vegetation Index
PUM	Product User Manual
PRU	Primary Reference Unit
PVR	Product Validation Report
SC	Specific Contract
SCL	Scene Classification Layer
SLIC	Simple Linear Iterative Clustering
STAC	SpatioTemporal Asset Catalogues
SU	Sampling Unit
TCD	Tree Cover Density
UTM	Universal Transverse Mercator
VHR	Very High Resolution
WMS	Web Map Service

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